

Google Maps API Key Setup Guide

Complete Step-by-Step Instructions

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Prerequisites

Before you begin, ensure you have:

- ☐ A Google account (Gmail)
- ☐ A valid credit/debit card (for billing verification - you won't be charged immediately)
- ☐ A web browser

Note: Google offers a **\$300 free trial credit** for new users and a **\$200 monthly credit** specifically for Google Maps Platform. Most development and learning activities will fall within these free limits.

Step 1: Create a Google Cloud Account

If you already have a Google Cloud account, skip to [Step 2](#).

1.1 Go to Google Cloud Console

Open your browser and navigate to:

```
https://console.cloud.google.com/
```

1.2 Sign In or Create Account

- If you have a Gmail account, sign in with your credentials
- If you don't have a Google account, click **Create account** and follow the prompts

1.3 Accept Terms of Service

- Read and accept the Google Cloud Platform Terms of Service
 - Select your country
 - Click **Agree and Continue**
-

Step 2: Create a Google Cloud Project

Every API key must belong to a Google Cloud Project. Think of a project as a container for your application's resources.

2.1 Go to Project Selector

Navigate to:

```
https://console.cloud.google.com/projectselector2/home/dashboard
```

Or click the project dropdown at the top of the Cloud Console (next to "Google Cloud").

2.2 Create New Project

1. Click **New Project** (or **Create Project** button)
2. Fill in the project details:
 - **Project name:** Enter a descriptive name (e.g., `maps-agent-project`)
 - **Project ID:** Auto-generated, but you can customize it (must be globally unique)
 - **Location:** Leave as "No organization" unless you're part of a Google Workspace organization
3. Click **Create**

2.3 Select Your Project

After creation:

1. Click the notification bell icon (top right) to see "Project created" notification
2. Click **Select Project** to switch to your new project

Or use the project dropdown and select your newly created project.

Verify: The project name should now appear in the top navigation bar.

Step 3: Enable Billing

Google Maps Platform APIs require a billing account. This is mandatory even for free tier usage.

3.1 Go to Billing Page

Navigate to:

```
https://console.cloud.google.com/billing
```

Or in the Console: **Navigation Menu (≡) → Billing**

3.2 Create Billing Account (First Time Users)

If you don't have a billing account:

1. Click **Create Account** or **Link a billing account**
2. Select your country and click **Confirm**
3. Choose account type:
 - **Individual** - For personal use
 - **Business** - For company use
4. Enter your payment details:
 - Name and address
 - Credit/debit card information
5. Click **Start my free trial** (if available) or **Submit and enable billing**

3.3 Link Billing Account to Project

If you have a billing account but it's not linked:

1. Go to **Navigation Menu (≡) → Billing**
2. Click **Link a billing account**
3. Select your billing account from the dropdown
4. Click **Set Account**

3.4 Verify Billing is Enabled

Navigate to:

```
https://console.cloud.google.com/billing/linkedaccount
```

You should see your project listed with a billing account linked.

Important: You won't be charged unless you exceed the free tier limits. Google provides \$300 in free credits for new accounts (valid for 90 days).

Step 4: Enable Required APIs

For this project, we need to enable **two APIs**:

API	Purpose
Directions API	Get directions between locations
Places API	Search for places (gas stations, restaurants, etc.)

4.1 Go to Maps API Library

Navigate to:

```
https://console.cloud.google.com/google/maps-apis/api-list
```

Or: **Navigation Menu (≡)** → **APIs & Services** → **Library** → Search for "Maps"

4.2 Enable Directions API

1. In the API Library, search for "**Directions API**"
2. Click on **Directions API** in the results
3. Click the **Enable** button
4. Wait for the API to be enabled (a few seconds)

Verify: The button should change from "Enable" to "Manage"

4.3 Enable Places API

1. Go back to the API Library (click **← APIs & Services** or navigate again)
2. Search for "**Places API**"
3. Click on **Places API** (or **Places API (New)**)
4. Click the **Enable** button
5. Wait for the API to be enabled

Note: "Places API (New)" is the latest version. Either version works for this project.

4.4 Verify Enabled APIs

To confirm both APIs are enabled:

1. Navigate to: **Navigation Menu (≡)** → **APIs & Services** → **Enabled APIs & Services**
2. You should see both:
 -  Directions API
 -  Places API

Step 5: Create an API Key

Now we'll create the API key that authenticates your requests.

5.1 Go to Credentials Page

Navigate to:

```
https://console.cloud.google.com/google/maps-apis/credentials
```

Or: **Navigation Menu (≡) → APIs & Services → Credentials**

5.2 Create API Key

- 1. Click **+ Create Credentials** at the top of the page
- 2. Select **API key** from the dropdown

5.3 Copy Your API Key

A dialog will appear showing your newly created API key:

API key created

Your API key is:
AIzaSyB_XXXXXXXXXXXXXXXXXXXXXXXXXXXX

Restrict your key to prevent unauthorized use in production.

- 1. **Copy the API key** immediately (click the copy icon or select and copy)
- 2. **Save it securely** - you'll need this for your application
- 3. Click **Close**

⚠ **Security Warning:** Never commit your API key to public repositories or share it publicly. Anyone with your key can use your quota and incur charges.

Step 6: Restrict Your API Key (Recommended)

Unrestricted API keys are a security risk. Follow these steps to restrict your key.

6.1 Access Key Settings

From the Credentials page:

- 1. Find your API key in the list under **API keys**
- 2. Click on the key name to open its settings

6.2 Set Application Restrictions

Under **Application restrictions**, choose based on your use case:

Restriction Type	Use Case
None	Development only (not recommended for production)
IP addresses	Server-side applications (web servers, scripts)

Restriction Type	Use Case
HTTP referrers	Client-side web applications
Android apps	Android mobile applications
iOS apps	iOS mobile applications

For this project (server-side Python script), choose:

1. Select **IP addresses (web servers, cron jobs, etc.)**
2. Click **Add**
3. Enter your server's IP address or **0.0.0.0/0** for development (allows all IPs - not for production!)

6.3 Set API Restrictions

Under **API restrictions**:

1. Select **Restrict key**
2. In the dropdown, select:
 - o ☒ **Directions API**
 - o ☒ **Places API**
3. This ensures the key can ONLY be used for these APIs

6.4 Save Restrictions

1. Click **Save** at the bottom of the page
2. Wait for the changes to propagate (may take a few minutes)

Best Practice: For production, create separate API keys for different applications and restrict each appropriately.

Step 7: Test Your API Key

Let's verify your API key works correctly.

7.1 Test Directions API

Open a browser and paste this URL (replace **YOUR_API_KEY** with your actual key):

```
https://maps.googleapis.com/maps/api/directions/json?
origin=Chennai&destination=Bangalore&key=YOUR_API_KEY
```

Expected Response: A JSON object with route information:

```
{
  "geocoded_waypoints": [...],
  "routes": [...],
```

```
"status": "OK"
}
```

7.2 Test Places API

Test with this URL (replace `YOUR_API_KEY`):

```
https://maps.googleapis.com/maps/api/place/nearbysearch/json?
location=13.0827,80.2707&radius=5000&keyword=restaurant&key=YOUR_API_KEY
```

Expected Response: A JSON object with nearby places:

```
{
  "html_attributions": [],
  "results": [...],
  "status": "OK"
}
```

7.3 Common Error Responses

Status	Meaning	Solution
OK	Success	API key is working
REQUEST_DENIED	API key invalid or restricted	Check key restrictions
OVER_QUERY_LIMIT	Quota exceeded	Check billing or wait
INVALID_REQUEST	Bad request parameters	Check your URL parameters

Step 8: Use the API Key in This Project

Now let's configure this project to use your API key.

8.1 Create Environment File

In the project root directory, create a file named `.env`:

```
# Create .env file
touch .env
```

8.2 Add Your API Keys

Open `.env` and add the following:

```
# Google Maps API Key (for Directions API and Places API)
MAPS_API_KEY=your_google_maps_api_key_here

# Google Gemini API Key (for the LLM - if using Google AI Studio)
GOOGLE_API_KEY=your_gemini_api_key_here
```

Replace:

- `your_google_maps_api_key_here` with the API key from Step 5
- `your_gemini_api_key_here` with your Gemini API key (get from [Google AI Studio](#))

8.3 Verify .env is in .gitignore

Ensure your API keys are not committed to version control:

```
# Check if .env is ignored
cat .gitignore | grep ".env"
```

If not present, add `.env` to your `.gitignore` file:

```
.env
```

8.4 Test the Agent

Run the Maps Agent to verify everything works:

```
# Using ADK CLI
adk run Maps_Agent

# Or using ADK Web Interface
adk web
```

Troubleshooting

API Key Not Working

Issue	Solution
"API key not valid"	Verify the key is copied correctly, no extra spaces
"API not enabled"	Go to API Library and enable the required APIs
"Billing not enabled"	Link a billing account to your project

Issue	Solution
"Quota exceeded"	Check your usage in Cloud Console
"Restricted API key"	Ensure the APIs are allowed in key restrictions

Check API Key Status

1. Go to **Credentials** page
2. Click on your API key
3. Check:
 - Status should be **Active**
 - API restrictions should include required APIs
 - Application restrictions should allow your IP

View API Usage

Navigate to:

```
https://console.cloud.google.com/google/maps-apis/metrics
```

This shows your API usage, errors, and latency.

Pricing & Free Tier

Google Cloud Free Trial

- **\$300 credit** for new users
- Valid for **90 days**
- Can be used for any Google Cloud service

Google Maps Platform Credits

- **\$200 monthly credit** for Maps Platform
- Applies to all Maps, Routes, and Places APIs
- Resets every month

API Pricing (after free credits)

API	Free Tier	Cost After
Directions API	~40,000 requests/month	\$5 per 1,000 requests
Places - Nearby Search	~5,000 requests/month	\$32 per 1,000 requests

For Learning/Development: The free tier is more than sufficient. You would need to make thousands of requests per day to exceed the free credits.

Set Budget Alerts

To avoid unexpected charges:

1. Go to **Billing** → **Budgets & alerts**
2. Click **Create Budget**
3. Set a monthly budget (e.g., \$10)
4. Configure alert thresholds (50%, 90%, 100%)
5. Add email notifications

Quick Reference

Important URLs

Purpose	URL
Cloud Console	https://console.cloud.google.com/
Project Selector	https://console.cloud.google.com/projectselector2/home/dashboard
Billing	https://console.cloud.google.com/billing
API Library	https://console.cloud.google.com/apis/library
Credentials	https://console.cloud.google.com/apis/credentials
Maps API Metrics	https://console.cloud.google.com/google/maps-apis/metrics
Gemini API Keys	https://aistudio.google.com/app/apikey

Required APIs for This Project

API	Service Name	Purpose
Directions API	directions-backend.googleapis.com	Get route directions
Places API	places-backend.googleapis.com	Search for places

Environment Variables

```
MAPS_API_KEY=AIzaSy...           # Google Maps API Key
GOOGLE_API_KEY=AIzaSy...         # Gemini API Key (for LLM)
```

Summary Checklist

Use this checklist to ensure you've completed all steps:

- ☐ Created Google Cloud account
- ☐ Created a new project
- ☐ Enabled billing for the project
- ☐ Enabled Directions API

- ☐ Enabled Places API
 - ☐ Created an API key
 - ☐ Restricted the API key (recommended)
 - ☐ Tested the API key with both APIs
 - ☐ Created `.env` file with API key
 - ☐ Verified `.env` is in `.gitignore`
 - ☐ Tested the Maps Agent
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